

Assignment 4 — MPI/OpenMP using MPJ Express (Complete Clean Setup)

This is the COMPLETE proper setup and execution flow from scratch in correct order.

You will use:

- Java
- Eclipse
- MPJ Express
- PowerShell

Goal:

```
Distributed Sum using MPI
```

PART 1 — DOWNLOAD MPJ EXPRESS

STEP 1 — DOWNLOAD MPJ EXPRESS

Download:

MPJ Express Download

STEP 2 — EXTRACT ZIP PROPERLY

After download:

- Right click ZIP
- Extract All

Extract directly to:

```
C:\mpj
```

IMPORTANT CHECK

After extraction:

OPEN:

C:\mpj

Inside it you MUST directly see:

bin
conf
doc
lib
src

VERY IMPORTANT:

conf

must exist here.

WRONG STRUCTURE

Do NOT keep:

C:\mpj\mpj-v0_44\mpj-v0_44

nested folders.

If nested:

- open inner folder
- cut all contents
- paste directly inside:

C:\mpj

PART 2 — SET ENVIRONMENT VARIABLES

STEP 3 — OPEN ENVIRONMENT VARIABLES

Press:

Windows Key

Search:

Environment Variables

Open:

Edit the system environment variables

Click:

Environment Variables

STEP 4 — CREATE MPJ_HOME

Under:

User Variables

Click:

New

Add:

Variable Name:

MPJ_HOME

Variable Value:

C:\mpj

Click OK.

STEP 5 — ADD BIN TO PATH

Still under:

User Variables

Find:

Path

Click:

Edit

Then:

New

Add:

C:\mpj\bin

Click:

OK → OK → OK

PART 3 — VERIFY INSTALLATION

STEP 6 — CLOSE OLD TERMINALS

Close:

- PowerShell
- CMD
- Eclipse

IMPORTANT.

STEP 7 — OPEN NEW POWERSHELL

Run:

```
echo $env:MPJ_HOME
```

Expected:

```
C:\mpj
```

STEP 8 — TEST MPJ

Run:

```
mpjrun.bat
```

If MPJ info appears:

☒ setup successful.

PART 4 — CREATE ECLIPSE PROJECT

STEP 9 — OPEN ECLIPSE

Create Java Project:

```
DS_Assignment4
```

STEP 10 — ADD mpj.jar

Right click project:

Build Path

→ Configure Build Path

Go:

Libraries

Click:

Add External JARs

Select:

C:\mpj\lib\mpj.jar

Apply and Close.

PART 5 — CREATE PROGRAM

STEP 11 — CREATE JAVA CLASS

Inside src create:

DistributedSum.java

STEP 12 — PASTE CODE

```
import mpi.*;

public class DistributedSum {

    public static void main(String[] args)
        throws Exception {

        MPI.Init(args);

        int rank =
            MPI.COMM_WORLD.Rank();
```

```
int size =
    MPI.COMM_WORLD.Size();

int[] numbers =
    {1,2,3,4,5,6,7,8,9,10};

int n = numbers.length;

int chunk = n / size;

int start = rank * chunk;

int end;

if(rank == size - 1) {

    end = n;

} else {

    end = start + chunk;
}

int localSum = 0;

for(int i = start;
    i < end;
    i++) {

    localSum += numbers[i];
}

int[] allSums =
    new int[size];

MPI.COMM_WORLD.Allgather(
    new int[]{localSum},
    0,
    1,
    MPI.INT,
    allSums,
    0,
    1,
    MPI.INT
);

int finalSum = 0;
```

```
if(rank == 0) {

    System.out.println(
        "Number of Processes: "
        + size
    );

    System.out.println(
        "\nIntermediate Sums:"
    );

    for(int i = 0;
        i < size;
        i++) {

        System.out.println(
            "Process " + i +
            " : " +
            allSums[i]
        );

        finalSum += allSums[i];
    }

    System.out.println(
        "\nFinal Sum = "
        + finalSum
    );
}

MPI.Finalize();
}
```

PART 6 — COMPILE PROGRAM

STEP 13 — SAVE FILE

CTRL + S

STEP 14 — OPEN POWERSHELL IN src

Go to:

```
C:\Users\HP\eclipse-workspace\DS_Assignment4\src
```

Open PowerShell there.

STEP 15 — COMPILE

Run:

```
javac -cp "C:\mpj\lib\mpj.jar" DistributedSum.java
```

VERIFY

Run:

```
dir
```

You should see:

```
DistributedSum.class
```

PART 7 — RUN PROGRAM

STEP 16 — EXECUTE MPI

Run:

```
mpjrun.bat -np 4 DistributedSum
```

EXPECTED OUTPUT

MPJ Express (0.44) is started in the multicore configuration

Number of Processes: 4

Intermediate Sums:

Process 0 : 3

Process 1 : 7

Process 2 : 11

Process 3 : 34

Final Sum = 55

IMPORTANT MPI FLOW

MPI.Init()

↓

Processes Created

↓

Array Divided

↓

Local Sums Calculated

↓

Allgather Combines Results

↓

Final Sum Displayed

↓

MPI.Finalize()

MOST COMMON ERRORS

package mpi does not exist

Cause:

mpj.jar path wrong

mpjexpress.conf not found

Cause:

MPJ_HOME wrong

ClassNotFoundException

Cause:

running from wrong folder

MOST IMPORTANT VIVA ANSWER

MPI (Message Passing Interface) is used in distributed systems to enable communication among multiple parallel processes. In this assignment, MPJ Express divides the array among multiple processes, computes local sums independently, and combines the partial sums to obtain the final result using distributed processing.

suppose i am using ubuntu linux terminal, so now tell me

Assignment 4 — MPI/OpenMP using Ubuntu Linux Terminal

Now we will do the assignment using:

- Ubuntu Linux
- Java

- MPJ Express
- Terminal

This is actually cleaner and easier than Windows.

WHAT YOU WILL DO

You will:

1. Install Java
2. Download MPJ Express
3. Set environment variables
4. Write MPI Java program
5. Compile
6. Run using multiple processes

Goal:

```
Distributed Sum using MPI
```

PART 1 — INSTALL JAVA

STEP 1 — UPDATE UBUNTU

Open terminal:

```
sudo apt update
```

STEP 2 — INSTALL JDK

Install Java:

```
sudo apt install openjdk-8-jdk
```

Press:

Y

when asked.

STEP 3 — VERIFY JAVA

Run:

```
java -version
```

Then:

```
javac -version
```

You should see version information.

PART 2 — DOWNLOAD MPJ EXPRESS

STEP 4 — DOWNLOAD MPJ

Go to Downloads folder:

```
cd ~/Downloads
```

Download MPJ:

```
wget https://sourceforge.net/projects/mpjexpress/files/mpj-v0_44.zip
```

STEP 5 — INSTALL unzip

```
sudo apt install unzip
```

STEP 6 — EXTRACT MPJ

```
unzip mpj-v0_44.zip
```

STEP 7 — MOVE MPJ TO HOME

```
mv mpj-v0_44 ~/mpj
```

STEP 8 — VERIFY STRUCTURE

Run:

```
ls ~/mpj
```

You should see:

```
bin
conf
lib
src
```

IMPORTANT:

```
conf
```

must exist.

PART 3 — SET ENVIRONMENT VARIABLES

STEP 9 — OPEN .bashrc

Run:

```
nano ~/.bashrc
```

STEP 10 — ADD THESE LINES AT BOTTOM

```
export MPJ_HOME=$HOME/mpj
export PATH=$PATH:$MPJ_HOME/bin
```

STEP 11 — SAVE FILE

Press:

```
CTRL + O
ENTER
CTRL + X
```

STEP 12 — APPLY CHANGES

Run:

```
source ~/.bashrc
```

STEP 13 — VERIFY MPJ_HOME

Run:

```
echo $MPJ_HOME
```

Expected:

```
/home/yourusername/mpj
```

STEP 14 — VERIFY MPJ

Run:

```
mpjrun.sh
```

If MPJ information appears:

☒ setup successful.

PART 4 — CREATE PROGRAM

STEP 15 — CREATE DIRECTORY

```
mkdir ~/DS_Assignment4
```

Go inside:


```
cd ~/DS_Assignment4
```

STEP 16 — CREATE JAVA FILE

```
nano DistributedSum.java
```

STEP 17 — PASTE CODE

```
import mpi.*;

public class DistributedSum {

    public static void main(String[] args)
        throws Exception {

        MPI.Init(args);

        int rank =
            MPI.COMM_WORLD.Rank();

        int size =
            MPI.COMM_WORLD.Size();

        int[] numbers =
            {1,2,3,4,5,6,7,8,9,10};

        int n = numbers.length;

        int chunk = n / size;

        int start = rank * chunk;

        int end;

        if(rank == size - 1) {

            end = n;
```

```
} else {

    end = start + chunk;
}

int localSum = 0;

for(int i = start;
    i < end;
    i++) {

    localSum += numbers[i];
}

int[] allSums =
    new int[size];

MPI.COMM_WORLD.Allgather(
    new int[]{localSum},
    0,
    1,
    MPI.INT,
    allSums,
    0,
    1,
    MPI.INT
);

int finalSum = 0;

if(rank == 0) {

    System.out.println(
        "Number of Processes: "
        + size
    );

    System.out.println(
        "\nIntermediate Sums:"
    );

    for(int i = 0;
        i < size;
        i++) {

        System.out.println(
            "Process " + i +
            " : " +
            allSums[i]
        );
    }
}
```

```
        );

        finalSum += allSums[i];
    }

    System.out.println(
        "\nFinal Sum = "
        + finalSum
    );
}

MPI.Finalize();
}
```

STEP 18 — SAVE FILE

Press:

```
CTRL + O
ENTER
CTRL + X
```

PART 5 — COMPILE PROGRAM

STEP 19 — COMPILE

Run:

```
javac -cp $MPJ_HOME/lib/mpj.jar DistributedSum.java
```

VERIFY

Run:

ls

You should see:

DistributedSum.class

PART 6 — RUN PROGRAM

STEP 20 — EXECUTE MPI

Run:

mpjrun.sh -np 4 DistributedSum

WHAT THIS MEANS

Part	Meaning
mpjrun.sh	starts MPI
-np 4	create 4 processes
DistributedSum	class name

EXPECTED OUTPUT

MPJ Express (0.44) is started in the multicore configuration

Number of Processes: 4



Intermediate Sums:

Process 0 : 3

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IMPORTANT MPI FLOW

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Processes Created

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Array Divided

↓

Local Sums Calculated

↓

Allgather Combines Results

↓

Final Sum Displayed

↓

MPI.Finalize()

MOST COMMON ERRORS

package mpi does not exist

Cause:

wrong mpj.jar path

mpjrun.sh command not found

Cause:

PATH not set

ClassNotFoundException

Cause:

running from wrong folder

MOST IMPORTANT VIVA ANSWER

MPI (Message Passing Interface) is used in distributed systems to enable communication among multiple parallel processes. In this assignment, MPJ Express divides the array among multiple processes, computes local sums independently, and combines the partial sums to obtain the final result using distributed processing.